AP Precalculus

Unit 3: Trigonometric and Polar Functions

Lesson 3.8 The Tangent Function

Name: ____________________________ Date: ________________ Period: _________

Learning Targets — I Can...

1. I can construct a representation of $f(\theta) = \tan\theta$ using the unit circle by computing $\sin\theta/\cos\theta$ and interpreting it as the slope of the terminal ray. *(LO 3.8.A)*
2. I can describe key characteristics of the tangent function: period π, location of vertical asymptotes at $\theta = \pi/2 + k\pi$, the function is always increasing, and the graph changes from concave down to concave up within each period. *(LO 3.8.B)*
3. I can describe the effect of additive and multiplicative transformations on the tangent function, including changes to period, phase shift, vertical dilation, and vertical translation. *(LO 3.8.C)*

What's in This Packet

#	**Component**	**Description**	**Score**
1	Warm-Up	Spiral review: unit circle values, slope, and asymptotes	________
2	Guided Notes	Tangent from the unit circle; period, asymptotes, transformations	________
3	Group Activity	Tiered tangent practice (R/A/E)	________
4	Exit Ticket	Compute $\tan\theta$; identify period and asymptote	________
5	Homework	Mixed practice + AP-style MC + preview of Lesson 3.9	________
		Total	________